

An Analytical Framework for Error and Hallucination in Deployed Generative AI Systems

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Abstract

Judging a response from a deployed generative AI system requires an applicable standard; explaining its production requires a different evidentiary basis. These requirements can diverge: information used in production may differ from the information that should govern evaluation, and a valid error classification may leave the production cause unresolved. We develop an authority-first framework that addresses these problems in sequence. The evaluator establishes the task and applicable authority policy, identifies assessable assertions and task obligations, and selects the governing standards. Assertions are classified as supported, contradicted, unsupported, or unresolved; obligations are assessed as satisfied, violated, or unresolved. These parallel comparisons support two related error classifications. Hallucination concerns a representational assertion contradicted by, or lacking support required from, its governing reference. Task-obligation fidelity error concerns failure to satisfy an applicable requirement. The classifications can overlap when a response both violates a requirement and misrepresents compliance or the requirement itself. We then distinguish identifying an error from locating its possible production source and attributing its cause. Applications to temporal change, continuing-project state, propagation, and retrieval-augmented generation show how the distinction guides evaluation of configured systems. Consequences inform the urgency of a response, while supported production findings inform the choice of controls. The framework supplies a conceptual sequence whose effects on evaluator agreement and control selection remain questions for empirical study.

Keywords: AI governance; AI evaluation; model error; hallucination; factuality; faithfulness; governing reference; task-obligation fidelity; project-state fidelity; RAG

1. Introduction

Suppose a generative AI system is asked to apply an organization’s current policy. Its accessible materials include a superseded manual and a later approved manual, but the response follows the older rule. The response may accurately reproduce the document used during production and still fail the task. To judge it, the evaluator must identify which policy governs. To explain it, the evaluator must investigate why the configured system delivered the older rule. Agreement with a production input answers neither question by itself.

We use *configured system* to mean the trained model together with the deployment resources, processes, settings, tools, and controls that help produce the response delivered to the user. Such a system may draw on organizational documents, retrieved information, prior interactions, or technical and procedural controls. Continuing work can also depend on decisions accepted during earlier interactions. The model is therefore one component of the system whose delivered response is being evaluated.

This setting creates two related problems. First, the information used to produce a response may differ from the information that should govern its evaluation. In the policy example, the older manual can be a production input while the currently approved manual supplies the evaluative standard. A source-faithful summary creates another possibility: the designated source governs what the summary should represent, even when general knowledge suggests a different account.

Second, identifying an error leaves open how the system produced it. The relevant policy may have been unavailable, omitted during context construction, supplied but insufficiently used during generation, or displaced by a later transformation. Several paths can produce the same delivered error. A single production failure can also lead to different outcomes: failed retrieval may be followed by abstention in one response and fabrication in another. Evaluation therefore needs to separate the response-level finding from the production investigation.

Existing work provides useful language for the response-level finding. Surveys and governance guidance discuss factual falsehood, lack of support, and inconsistency with relevant material ([1]–[3]). Research on summarization and source fidelity distinguishes faithfulness to supplied material from external factuality ([4], [5]). Benchmark and survey work also examines distinctions among contradiction, unsupported content, and the knowledge against which a claim is assessed ([6], [7]). These distinctions help characterize a discrepancy once the applicable reference is established. Our starting question is how to establish that reference for the assertion and task being evaluated.

We call the resulting approach *authority-first evaluation*. Here, authority means the standing of a reference or requirement to govern a particular evaluation, as established by the task and applicable priority rules. It is more specific than general credibility. A historical manual can govern a question about an earlier rule; a current manual can govern a question about the rule now in force. Selecting the governing standard makes the evaluative question explicit before a discrepancy is classified.

The framework uses two parallel comparisons. One concerns what the response represents: its assertions are compared with governing references. The other concerns what the response was required to do: it is compared with applicable task obligations. We use *task-obligation fidelity* for conformity with requirements established by a governing source, instruction, contextual record, or authorized project state. Authorized project state consists of accepted decisions, requirements, and statuses governing continuing work at the time relevant to the task. An approved requirement can govern that work; an unapproved proposal remains a proposal.

These comparisons allow hallucination and task-obligation fidelity error to be assessed separately or together. A response can exceed a word limit without making a false claim. It can also exceed the limit while claiming compliance. The distinction keeps a behavioral failure visible while allowing a misrepresentation of compliance to be evaluated as an assertion.

We develop this conceptual analysis in the order needed to answer four questions: What governs the response? What discrepancy occurred? What can the evidence establish about its production? What response is appropriate? Section 2 first describes production so that its inputs and outputs can be distinguished from evaluative authority. Sections 3 and 4 establish governing standards and error

classifications. Section 5 develops the separate production inquiry, Section 6 applies the distinctions, and Section 7 connects the findings to consequences and controls.

2. Response Production in a Deployed System

2.1 Accessible information and operative context

To separate evaluation from production, we first identify what can enter a response and what the system actually delivers. At interaction time t , let x_t denote the current production input, including the user’s request and applicable instructions supplied with it. Let $\mathcal{H}_{t-1}$ denote prior interaction history, and let E_t denote potentially accessible external resources, such as documents, databases, tools, and current observations.

Accessible material becomes useful for generation through a context-construction process. This process may retrieve, select, compress, order, or label information. Denoting it by R , we represent the supplementary context assembled for generation as:

$$C_t = R(x_t, \mathcal{H}_{t-1}, E_t). \quad (1)$$

We call C_t the *operative context*: the supplementary information actually presented for generation. Equation (1) distinguishes that context from all potentially accessible material. A required passage may exist in a repository yet remain outside the operative context because it was not retrieved, was displaced, or lost a qualification during compression.

Inclusion and use also differ. Research on long contexts shows that performance can depend on the position of relevant information within the supplied material ([8]). A production investigation therefore needs to examine both what entered the context and how the generation process used it. In the policy example, obtaining the approved manual is one condition; retaining its approval status and relevant rule in the operative context is another.

The current input x_t has two production roles. It guides the assembly of supplementary material in Equation (1), and it enters generation directly in Equation (2). History and external material are represented here through context construction. This separation helps distinguish a deficient request or instruction from missing supplementary information. Section 3 will introduce a different object, the evaluative task specification, to state what the response was required to do.

2.2 Candidate response and delivery

Once the operative context has been assembled, the deployed generation process produces a candidate response. Let $M(t_0)$ denote the trained model, with t_0 marking its training-information boundary in simplified form. Training material can span different dates; the marker identifies the information boundary relevant to the analysis rather than one exact date shared by every item. Let λ denote the operative decoding procedure and settings. The distribution over candidate responses can then be written as:

$$\tilde{y}_t \sim P_{M(t_0),\lambda}(\cdot | x_t, C_t). \quad (2)$$

In Equation (2), $\tilde{y}_t$ is the candidate response and $P_{M(t_0),\lambda}$ summarizes the deployed generation distribution conditional on the input and context. The model supplies conditional predictions, while the decoding procedure and successive selections realize a response. Including λ makes explicit that the generation process depends on deployment choices as well as on the trained model.

This distinction matters when information changes after training. A model’s prior information may be outdated, but current context or tools can supply an update. Whether the update affects the candidate response depends on its availability and use. The training boundary identifies one condition to investigate; the response must still be examined as the outcome of the configured production process.

The candidate response can undergo further processing before delivery. Let Σ denote the applicable post-generation rules and controls, and let I_Σ denote the process implementing them. A compact representation is:

$$y_t = I_\Sigma(\tilde{y}_t, x_t, C_t, E_t). \quad (3)$$

Here y_t records the delivered response or a withheld-response outcome. The inclusion of E_t allows a verification process to consult external material beyond the context supplied during generation. As in Equation (1), availability is distinct from actual use: the production record must identify what a verifier consulted. A process can verify, qualify, redact, rewrite, replace a candidate with a refusal, or block delivery.

The distinction between candidate and delivered response becomes material when a later process removes a qualification, inserts an explanation, or prevents an erroneous claim from reaching the user. If delivery is withheld, the assertion branch evaluates only claims actually communicated, such as claims in a refusal notice. The task-obligation branch can separately assess whether withholding satisfied the applicable requirements.

Equations (1)–(3) are schematic descriptions of production stages. They organize information access, candidate generation, and delivery; their mappings stand for operations under the recorded configuration. A deployment may iterate through retrieval, tool use, and generation before producing the final candidate. The representation gives an investigation distinct objects to examine without assuming that every implementation follows one simple pass.

We can now state the first evaluative problem more precisely. The operative context explains part of what was supplied for production. It acquires governing authority only if the task and applicable priority rules give it that standing. We turn next to how that standing is established.

3. Governing References and Response Evaluation

3.1 Establishing the task and selecting authority

A delivered response can contain several claims that require different standards. Following the claim-level approach used in factuality evaluation ([9]), we call an *assertion* an explicit or reasonably implied claim whose relation to an applicable reference can be assessed. The assertion should preserve the

response’s meaning and qualifications. For example, “the report states that the deadline is Friday” represents the report’s contents; “the deadline is Friday” may assert the operative deadline itself.

Evaluation begins by establishing what the response was required to do. Let q_t denote the *evaluative task specification*, reconstructed from x_t and any controlling instructions, authorized contextual records, or project state applicable to the response. The distinction is practical: x_t records what entered production, whereas q_t states the evaluative requirements. If a controlling decision failed to enter production, it can still govern evaluation.

Let Π_t denote the *authority policy*: the rules assigning applicability and priority among candidate standards for the task. In a source-faithful summary, that policy may designate the supplied source as governing its representation. In a current-policy task, it may give the currently approved manual priority over superseded material. Establishing these rules before choosing a preferred answer makes the standard visible and reduces opportunities for a convenient retrospective judgment.

Return to the opening example. The available materials include an older manual, a later approved manual, and a user message quoting the older rule. The task specification establishes that the response must apply the current policy. Provenance, effective dates, and approval status distinguish the materials. The authority policy then selects the currently applicable approved manual. If the task instead asks what the earlier manual required, the temporal target changes and the earlier manual may govern. The standard follows the task rather than the mere presence or recency of a document.

To represent this selection, let a_{tj} denote assertion j in the delivered response, where j indexes the assertions identified for evaluation. Let k_{tj} denote the assertion’s evaluative role, such as a claim about an external fact or a designated source. Let $\mathcal{A}_{tj}$ be the set of potentially applicable candidate references, and let ρ be the selection rule applying the task and authority policy. When these inputs establish a governing reference, we write:

$$r_{tj}^* = \rho(a_{tj}, q_t, k_{tj}, \mathcal{A}_{tj}; \Pi_t). \quad (4)$$

The star in r_{tj}^* marks the selected reference. Equation (4) states the dependency of selection; it is conditional on a sufficiently specified task and authority policy. If a relevant priority rule is missing or materially contested, the evaluator should record that conflict and resolve or explicitly stipulate the rule before making a classification that depends on it. Selecting an arbitrary reference would conceal the unsettled evaluative premise.

A governing reference may consist of a record or a defined set of materials needed to assess the assertion. For a claim about what an instruction required, the applicable instruction may suffice. For a claim of compliance, the governing basis may also require the actual response and an agreed measurement convention. The reference supplies the relevant basis for comparison; it need not have been received by the model.

Table 1 distinguishes several assertion roles to show why the same response may require more than one governing reference.

Table 1. Assertion Roles and Governing References

Assertion role	Potential governing reference	Illustration
External factual claim	Verifiable information applicable to the task’s temporal target	Who held an office on a specified date; a price or rule at the relevant time
Designated-source claim	The designated source, with its provenance and qualifications	What a supplied paper states
Claim about an applicable instruction	The controlling instruction or requirement	Whether the task specified a scope, format, or prohibited transformation
Contextual claim	The authoritative contextual record	What was established earlier in an exchange
Project-state claim	The authorized project state	Whether a proposal was accepted, revised, rejected, or superseded

These are evaluative roles rather than mutually exclusive claim types. If one assertion performs several roles, each relevant relation should be recorded separately unless the authority policy establishes which role governs the task. This preserves the difference between an accurate account of a source and an accurate assertion about the world. It also prevents external knowledge from silently replacing a designated source in a task whose purpose is to represent that source.

3.2 Comparing an assertion with its governing reference

Selection establishes the standard; comparison establishes how the assertion relates to it. Let v denote the comparison rule, with q_t and Π_t determining what support the task requires. Let V_{tj} denote the resulting classification:

$$V_{tj} = v(a_{tj}, r_{tj}^*; q_t, \Pi_t). \quad (5)$$

In Equation (5), the possible classifications are *supported*, *contradicted*, *unsupported*, and *unresolved*. An assertion is supported when the reference supplies the support required for that claim under the task. It is contradicted when the governing evidence conflicts with the claim. Where an affirmative conflict is established, contradiction is the more specific finding than mere lack of support.

Unsupported applies when the governing reference is sufficiently identified and available for the evaluator to determine that required support is absent. Suppose a task requires a faithful summary of a designated report, but the response adds a numerical result for which the complete relevant report supplies no basis. The addition is unsupported under that task, even if the same number might be true in another setting. The finding concerns the required evidentiary relationship; absence of support is not itself proof of external falsehood.

Unresolved applies when the available evidence prevents a decision among the other classifications. If the report’s missing appendix could supply the numerical result, the evaluator cannot yet establish the absence of required support. Ambiguous claims or conflicting evidence can produce the same limit. The record should identify what remains missing or unsettled. An unresolved relation establishes neither an error nor a passing result.

This distinction also identifies the next evaluative step. A demonstrably unsupported addition can be classified against the designated source. An unresolved addition may first require recovering the appendix, clarifying the claim, or resolving an evidentiary conflict. Recording uncertainty in that form preserves what the evaluation has established and what it still needs.

3.3 Assessing task obligations in parallel

Some requirements concern the response’s behavior or form rather than an assertion. A 200-word limit can be violated by an otherwise accurate summary. Evaluating that violation requires comparing the delivered response with the applicable length requirement, regardless of whether the response claims compliance.

The evaluator therefore identifies each task obligation from q_t under Π_t , establishes its governing requirement, and records whether the response satisfies it, violates it, or leaves the result unresolved. Relevant conventions must be stated where they affect the outcome. A disputed counting convention can matter near a word limit; an unambiguous excess can establish the violation directly.

The two branches answer complementary questions. Assertion-reference comparison concerns what the response represents. Obligation assessment concerns whether the response meets what the task requires, including source fidelity when that is an obligation. Their relationship becomes clear when a response both fails a requirement and makes an inaccurate claim about it. We develop that overlap next.

4. Hallucination and Task-Obligation Fidelity

We use *hallucination* for a representational assertion in a delivered response that is contradicted by, or lacks support required from, its governing reference. The assertion can concern an external fact, a designated source, an instruction, a contextual record, or authorized project state. This is our proposed umbrella usage. The selected reference and the task’s support requirement determine how it applies.

A *task-obligation fidelity error* occurs when the delivered response violates an applicable requirement. That requirement can concern response form, such as length, or substantive conduct, such as faithful representation of a designated source or continuation from an approved project decision. The violation can occur through what the system does even if the response makes no inaccurate assertion about that conduct.

Consider a source-faithful summary required to contain no more than 200 words. An accurate 300-word summary violates the length obligation. If it makes no claim about its length or compliance, that excess is a task-obligation fidelity error. Now suppose the response also says, “This summary meets the 200-word limit.” The instruction establishes the limit, and the actual count establishes that the statement of compliance is false. The response contains both an obligation violation and a hallucination concerning compliance. The instruction alone establishes the requirement; the instruction together with the response count establishes the inaccurate claim.

A source-fidelity failure can also enter both branches. If the summary attributes an invented numerical result to its designated report, the assertion lacks the required support and the response violates its source-

fidelity obligation. The branches can therefore overlap without becoming interchangeable. One records the assertion’s relationship to a reference; the other records the response’s conformity with a requirement.

Domain-specific labels can retain their value within this sequence. An evaluator may use factual error, source-unfaithful claim, instruction misrepresentation, or project-state error to identify the particular domain. The authority-first comparison remains applicable even when a narrower convention reserves the term hallucination for only some of those cases. Explicitly recording the assertion, reference, and relation makes the evaluative finding intelligible across naming conventions.

The classification establishes an outcome at the level of the delivered response. It leaves a separate production question. A missing retrieval result followed by abstention contains no discrepant assertion on those facts alone, although abstention may or may not satisfy the task. A fabricated answer following the same retrieval failure can contain a hallucination once the governing reference establishes the discrepancy. The production condition and the response-level error must therefore be investigated as distinct objects.

5. Localization and Causal Attribution

5.1 From an identified error to possible production paths

After identifying an error, we can ask where it could have arisen and what evidence establishes how it arose. We use *localization* for identifying a plausible production locus or set of loci. We use *causal attribution* for establishing that a condition or process at a locus contributed to the delivered error. Localization narrows an investigation; attribution supports an explanation.

The production description in Section 2 supplies concrete places to examine: the trained model, context construction, generation and decoding, and post-generation processing. To relate those places to broader processes, we use three mechanism families developed in a companion paper ([10]): Model Approximation, Autoregressive Realization, and Deployment Mediation. We introduce them here for their diagnostic role rather than requiring the reader to consult that paper before applying the evaluation sequence.

Model Approximation concerns the formation and limitations of the finite trained model, $M(t_0)$, from bounded information through design-contingent choices. Autoregressive Realization concerns how conditional predictions, decoding, and earlier selections shape the candidate response in Equation (2). Deployment Mediation concerns how resources, tools, interaction history, context construction, and later controls affect production in Equations (1) and (3). The families can interact: omitted current information may leave an outdated model association salient, while decoding or verification affects whether that association reaches delivery. Evidence from knowledge-grounded conversation indicates that both dataset construction and model behavior can contribute to hallucination ([11]), reinforcing the need to investigate more than one possible production locus.

A discrepancy about the approved policy illustrates the resulting investigation. The approved passage may not have been retrieved. It may have been retrieved but omitted or stripped of its approval status during context construction. It may have entered the operative context while the candidate response followed the

superseded rule. Alternatively, the candidate may have used the approved rule correctly before a later process altered it. Each path can lead to the same delivered claim, but the relevant evidence and possible control differ.

The response and approved policy can establish the evaluative discrepancy. Distinguishing these production paths requires records beyond that comparison. This is the evidentiary boundary between identifying the error and explaining its production.

5.2 Evidence for localization and attribution

Retrieval logs can establish what a search returned. Operative-context records can establish whether the relevant passage entered generation and whether provenance or qualifications survived. Stored candidate and delivered responses can establish whether a post-generation process changed the assertion. Repeated trials can reveal variation under recorded conditions, and controlled interventions can test whether changing a suspected production condition changes the outcome.

These forms of evidence support different strengths of inference. Observing that a passage was absent identifies a production condition. Establishing its contribution to the error may require comparisons that isolate the relevant change or other evidence of the production dependency. A candidate-to-delivered comparison can directly locate the introduction of a discrepancy during transformation, while further evidence may be needed to explain why the transformation occurred. Incomplete records can leave several plausible paths open even after the response-level error is settled.

Table 2 applies the distinction to illustrative cases. Each classification assumes that the task and governing reference establish the stated response-level finding. The production inquiries are nonexclusive, and the listed records indicate evidence needed to investigate contribution rather than causes inferred from the error label.

Table 2. Response-Level Findings and Production Evidence

Observed condition	Response-level finding	Production inquiry and relevant evidence
Retrieval returns no relevant material and the system abstains without making a discrepant claim	No hallucination on these facts alone; obligation assessment depends on the task	Retrieval logs establish the missing result. If abstention violates the task, explaining how that outcome arose requires a separate production inquiry.
Required source material is not retrieved, and the response invents an answer attributed to the designated governing source	Hallucination involving the source; a source-fidelity obligation may also be violated	Retrieval, context, and generation records help test how absent material and realization contributed.
The governing source is accurately represented in the operative context, but the delivered response contradicts it	Hallucination involving the source	Candidate and delivered records distinguish generation from later transformation. Trials and controlled changes can investigate model and realization contributions.
A purportedly current factual answer conflicts with the verified reference for the specified time	Factual hallucination	Training-information boundaries, context records, tool traces, and controlled comparisons help distinguish outdated prior information, absent updates, realization, and later mediation.
A post-generation process inserts an explanation lacking support required from the governing reference	Hallucination introduced during post-generation processing	Candidate-to-delivered comparison establishes the transformation. Verification inputs, rules, and process records help explain its production.

The table also shows why a cause cannot be read directly from an outcome category. Several production paths can support the same classification, and a single failed condition can have different downstream effects. The next applications examine settings in which time, retained state, and retrieval make these differences especially visible.

6. Applications of the Framework

6.1 Temporal targets and changing information

The interaction time t identifies when the response is produced. The time governing an assertion follows the task. A question may concern a current officeholder, a historical price, or a rule in force on an earlier date. Evaluation must therefore preserve the temporal target in q_t and select the reference applicable to that target. An assertion accurate at one time may be contradicted by the standard governing a different time.

Consider a model trained on information available through February. On May 31, a user asks a deployed system for a stock’s closing price that day, after the close. Suppose the system has no access to

subsequent market information but supplies a price as the requested closing price. If the answer differs from the verified May 31 closing price, the assertion is contradicted by its governing reference and can be classified as a hallucination under our usage.

The temporal setup limits the production conclusion. The May 31 price was unavailable at the training boundary, so this finding cannot establish a failure to learn that future price during training. The investigation instead concerns how the configured system responded to a request requiring information beyond its available record. Access, recognition of the temporal requirement, generation, and later verification or abstention are relevant conditions to examine. The evaluative finding can be firm while the production explanation remains open.

6.2 Authorized project state and propagation

Continuing work introduces another changing reference: the authorized project state. Suppose a team discusses one proposal, later rejects it, and approves a different course. Both discussions remain accessible. When the system is asked to continue from the approved decision, the accepted course governs the task. Mere retention of the rejected proposal gives it no renewed authority.

A response that continues according to the rejected proposal violates the project-state obligation. If it also states that this proposal remains approved, the assertion misrepresents authorized project state and is contradicted by the decision record. The evaluator therefore needs approval, revision, and supersession information alongside the interaction time. The production investigation can then examine whether the current state was accessible, correctly selected, and effectively used.

Propagation concerns a different relationship: how an earlier error enters later production. Suppose the actual deadline is Friday, but a response states Monday. The system retains that incorrect date and later retrieves it to prepare a schedule. The Friday deadline still governs evaluation of the later assertion. The retained Monday record is a possible production input whose authority must be assessed rather than assumed.

Evidence of propagation may include the initial error, the record retaining it, its retrieval, and the later response's use of that record. Reappearance of the same wrong date alone may leave independent production paths possible. A supported propagation account requires evidence connecting the episodes. This separates the evaluative persistence of the discrepancy from the production route through which it may have spread.

6.3 Retrieval and context construction

Retrieval-augmented generation, or RAG, supplies external information to generation through retrieval and contextual integration ([12]). It can make current, private, or task-specific information available to the model. It also creates an observable sequence through which available material may be selected, transformed, and used.

A typical deployment constructs a query, searches a corpus, ranks material, and places selected content in the operative context. The query can misstate the information need, retrieval or ranking can miss the applicable source, and context construction can remove provenance, temporal qualifications, or authority

labels. Prior work identifies failures across these pipeline stages ([13], [14]). Each stage therefore presents a different question for production investigation.

Retrieving the approved manual is one step in the opening example. Including its relevant passage and approval status is another. Using that passage to produce the candidate response is a further step, followed by any processing before delivery. A successful retrieval event cannot establish success at all later stages. Records that distinguish the stages help locate where the discrepancy was introduced or left uncorrected.

The governing-reference question remains separate throughout. Retrieved material may govern the task, conflict with the governing reference, or be irrelevant to the assertion. A response agreeing with an obsolete retrieved manual can still fail a current-policy task. Conversely, a response contradicting a retrieved document may be justified when the task requires a different authoritative standard. The evaluator applies q_t and Π_t before treating agreement or disagreement with retrieval as an error finding.

RAG thus changes the information entering production and the evidence available for investigation. Authority-first evaluation determines the standard against which the delivered response is judged. Keeping these roles distinct allows retrieval improvements to be evaluated for the particular problem they are intended to address.

7. Discussion and Conclusion

7.1 Consequences and control selection

Classification and production analysis leave a further governance question: what should be done about the finding? An error’s occurrence does not determine its consequence. Reliance, retention, and propagation can make the same false statement consequential in different ways. An inaccurate date ignored in a casual exchange may have little effect. Entering it into an operational schedule can expose subsequent work to the same discrepancy.

Consequence and production findings therefore inform different aspects of the response. Actual or potential consequences inform priority and urgency. Supported production findings inform which controls are plausible matches for the problem. Evidence that required information never entered the operative context can support investigating retrieval or access changes. Evidence that an accurate candidate was altered before delivery can support investigating the transformation process. These are diagnostic matches whose effectiveness requires evaluation in the deployment.

Urgent containment can proceed while causal attribution remains incomplete. An organization can correct a retained deadline, suspend reliance on a disputed response, or obtain authoritative verification before completing the production inquiry. A durable control decision should then use the evidence available about how the error occurred. This keeps immediate protection compatible with a more demanding standard for causal claims.

Controls also interact. Retrieval can supply an update while generation still follows an older association; verification can prevent delivery of an error introduced earlier. A change aimed at one stage should

therefore be assessed through the resulting configured system and task. The object of governance is the production arrangement that delivers the response, including the interactions among its components.

7.2 An evaluation sequence

The preceding distinctions can be brought together in an eight-step sequence:

1. Specify the task, including its temporal target, and the applicable authority policy.
2. Identify assessable assertions and applicable task obligations.
3. Select each governing reference and requirement, recording material authority conflicts.
4. Classify assertion-reference relations as supported, contradicted, unsupported, or unresolved, and obligations as satisfied, violated, or unresolved.
5. Determine whether the findings establish hallucination, task-obligation fidelity error, or both.
6. Identify plausible production loci and make causal claims to the extent supported by production evidence.
7. Assess consequences, including reliance, retention, and possible propagation.
8. Select and evaluate controls appropriate to the supported findings and deployment context.

This is an analytical order rather than a requirement to delay action until every inquiry is complete. Task and authority specification establish the basis for classification. The classified discrepancy identifies what a production explanation must explain. Consequence assessment establishes the reason and urgency for intervention, while production evidence helps direct durable controls. In practice, those later inquiries can proceed together, and immediate containment can precede causal resolution.

7.3 Scope and empirical development

The framework makes an authority choice explicit but leaves its substantive adoption to the responsible evaluator or organization. Classifications are conditional on that choice. Where the task or priority rule is contested, an evaluation should expose the disagreement and show which findings depend on it. This makes the premises reviewable rather than resolving every authority dispute through the framework itself.

The remaining development concerns evidence and use. Empirical studies can test agreement on reference selection and discrepancy classification under specified policies. For example, an evaluator study could compare agreement under ordinary review with agreement after participants receive the same task specification, authority policy, and classification sequence. Instrumented deployments can examine whether production records and controlled changes improve localization or support causal conclusions. Control studies can then evaluate whether interventions matched to those findings reduce the relevant errors under comparable tasks and configurations.

The analysis separates consequences from occurrence while leaving the measurement of adverse effects dependent on the application. It also allows reliable response-level classification in systems whose production history remains inaccessible. Closed or weakly instrumented deployments may support a firm

evaluative finding and a limited production explanation. Keeping that boundary explicit is part of a useful evaluation record.

7.4 Conclusion

A response is produced from inputs supplied to a configured system and evaluated against standards applicable to its task. Those inputs and standards can differ. The opening policy example illustrates the consequence: accurately following an obsolete production input can still yield an incorrect response to a current-policy task.

Authority-first evaluation addresses this problem by establishing the task and governing standards before classifying discrepancies. Comparing assertions with references and responses with obligations preserves two related findings: hallucination and task-obligation fidelity error. Their overlap can be recorded without treating every behavioral failure as a representational claim.

A valid classification then creates a production question rather than answering it. Records and interventions determine how far an evaluator can go in locating the error and attributing its cause. Consequences inform the urgency of action, and supported production findings inform the controls worth investigating. The resulting sequence connects judgment, explanation, and response while preserving the evidence required for each. Whether it improves evaluator agreement and control selection remains an empirical question.

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